**Yoni Nachmany**
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**Research Engineer** building foundation models, multidimensional data pipelines, and multimodal AI (CV x NLP).
**Keywords**: deep learning, ML weather prediction, digital twins, multilingual AI, crowdsourcing, medical imagery.

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## Experience

**Zeus AI** — Remote
**Head of Data Engineering, First Hire** — May 2024 – May 2025

- Operationalized EarthNet PyTorch foundation model for gap-filled global atmospheric weather 50x faster than traditional systems based on Apple's 4M (Massively Multimodal Masked Modeling) applied to 4D sensor data.
- Architected live ingestion/inference pipelines with AWS S3/SNS/Lambda and H100s on Modal/Lambda Labs.
- Scraped, filtered, normalized, and tokenized TBs of video-like satellite imagery with tens of variables on HPC.
- Deployed web demo of hourly changes in solar potential for energy users, visualizing Zarr arrays in the browser.

**The Linux Foundation** — Contract, Remote
**Senior Geospatial Data Scientist, Data Pipeline Task Force** — March 2024 – May 2024

- Prototyped Parquet changelogs with PySpark/Databricks for map data used by Microsoft, Meta, and Amazon.

**Gro Intelligence** — New York City
**Climate Data Scientist, Data Platform Project Lead** — April 2023 – March 2024

- Orchestrated 33 Airflow DAGs to compute 100+ years of past/projected indices in a versioned array data lake.
- Implemented and oversaw development of internal libraries to standardize Xarray metadata and input/output.

**The New York Times** — Contract, New York City
**R&D Engineer, Spatial Journalism** — December 2021 – September 2022

- Scaled transcription of scanned newspapers with multimodal APIs and evaluation framework for 10M articles.
- Implemented 3D satellite stereo reconstruction with photogrammetry libraries to support newsroom coverage.

**Mapbox** — Washington, D.C.
**Geospatial Software Engineer** — September 2019 – November 2021

- Updated 135M $km^2$ of satellite imagery for 3D map launch and saved $1M+ by leveraging open aerial imagery.
- Improved the Mapbox Satellite UX for all users loading data across zoom levels groups with novel data fusion.
- Implemented and evaluated deep learning-based object detection/super-resolution (EDSR) for map customers.

## Research

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|---|---|
| *Global Atmospheric Data Assimilation with Multi-Modal Masked Autoencoders* | Preprint 2024 |
| *Learning Word Ratings for Empathy and Distress from Document-Level User Responses* | LREC 2020 |
| *Design Considerations for High Impact, Automated Echocardiogram Analysis* | ICML 2020 |
| *Mitigating Geographic Bias of Image Classifiers with Multilingual Image Data* | D4GX 2019 |
| *Detecting Roads from Satellite Imagery in the Developing World* | CVPR 2019 |
| *Generating a Training Dataset for Land Cover Classification to Advance Global Development* | NeurIPS 2018 |
| *Decoded: Crowdsourcing Summaries of Internet Privacy Policies* | HCOMP 2017 |

## Talks

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|---|---|
| *Generating Elevation Data from Stereo Pairs of Satellite Imagery* | GeoPython 2023 |
| *Depth Perceptions - Perspectives on 3D & Digital Twins* | SatSummit 2022 |
| *Under-Mapped Spaces: New Methods and Tools for Critical Storytelling* | Stanford 2022 |
| *Deep Transfer Learning for Land Cover Classification on Open Multispectral Satellite Imagery* | FOSS4G 2019 |

## Education

**University of Pennsylvania** — Philadelphia, PA
MSE, Data Science — May 2019
BSE, Networked and Social Systems Engineering — May 2018